

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Semester – II of B. Pharm. Examination

University Backlog Theory Examination October/November 2016

PH119 Pharmaceutical Chemistry-II(Organic)

Date: 21/10/2016, Friday

Time:10:00 a.m To 01:00 p.m

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION-I

Q.1 Attempt all questions. Each question is of one marks

20

- 1 The bond angle of beryllium chloride is
[A] 120°
[B] 180°
[C] 170°
[D] 110°
- 2 Which of the following molecule having sp^2 hybridization?
[A] BF_3
[B] BeCl_2
[C] CaCl_2
[D] CH_4
- 3 The amount of energy liberated or consumed, when a bond is broken or formed is known as-
[A] Bond dissociation energy
[B] Energy of activation
[C] Energy maxima
[D] Exothermic energy
- 4 Relative number of the different kinds of atoms in a molecule known as-
[A] Molecular formula
[B] Empirical formula
[C] Gibb's formula
[D] Atomic formula
- 5 "The greater the number of alkyl groups attached to the doubly bonded carbon atoms, the more stable the alkene"-
[A] Saytzeff's orientation
[B] Anti Markovnikov addition
[C] Markovnikov addition
[D] None of above
- 6 Reactions that from the stand point of orientation give exclusively or nearly exclusively one of several Isomeric products are called-
[A] Heterolytic reaction
[B] Stereospecific reaction
[C] Homolytic reaction
[D] Regioselective reaction

- 7 Which of the following is not an example of intermolecular forces?
 - [A] Dipole-dipole interaction
 - [B] Hydrogen bonding
 - [C] Repulsive forces
 - [D] Vander Waals forces
- 8 Anti-Markovnikov addition of HBr on propylene yields-
 - [A] *neo*-pentyl bromide
 - [B] *n*-propyl bromide
 - [C] 2-bromo propane
 - [D] *iso*-propyl bromide
- 9 A reaction in which two molecules combine to yield a single molecule of product is called –
 - [A] Heterolytic reaction
 - [B] Addition reaction
 - [C] Homolytic reaction
 - [D] Substitution reaction
- 10 Which of the following is not an example of organometallic compound?
 - [A] R-MgX
 - [B] R_2CuLi
 - [C] LiAlH_4
 - [D] N_2H_4
- 11 A primary alkyl halide would preferred to undergo _____.
 - [A] S_N^1 Reaction
 - [B] S_N^2 Reaction
 - [C] α -Elimination
 - [D] Realization
- 12 General formula for 2 amino carboxylic acid is
 - [A] $\text{R-CH(NH}_2\text{)COOH}$
 - [B] R-CHCOOH
 - [C] $\text{R-CH(NH}_2\text{)}$
 - [D] R-CH
- 13 The molecular formula of _____ fit the general formula $(\text{C}_n\text{H}_{2n-2})$
 - [A] Alkynes
 - [B] Alcohol
 - [C] Alkenes
 - [D] Alkanes
- 14 Which compound is secondary alcohol?
 - [A] Butane – 1-ol
 - [B] Propane – 2-ol
 - [C] Isopropyl alcohol
 - [D] 2-Methylpropane-2-ol
- 15 Amines which are bonded in Alkyl group are:
 - [A] primary amines
 - [B] secondary amines
 - [C] tertiary amines
 - [D] quaternary amines
- 16 CH_3CONH_2 is formula for:
 - [A] amine
 - [B] alkyl amine
 - [C] methanamide
 - [D] ethanamide

- 17 Reduction of nitrobenzene will produce:
 [A] phenyl amine
 [B] water
 [C] aniline
 [D] both A and B
- 18 Toluene react with Halogen in the presence of iron(III) chloride giving ortho and para halo compounds. The reaction is:
 [A] Electrophilic elimination reaction
 [B] Electrophilic substitution reaction
 [c] Free radical reaction
 [D] Nucleophilic substitution reaction
- 19 Alkanes whose one or more atoms are replaced by halogen atoms are called
 [A] halogens
 [B] Halogenoalkanes
 [C] Halogenoalkenes
 [D] carbonyl
- 20 Formula of carbonic acid is
 [A] HCO_2
 [B] HCHO_3
 [C] H_2CO_2
 [D] H_2CO_3

SECTION-II

- Q.2** Attempt any **TWO** of the following
- A Write reaction, mechanism, reactivity and orientation for SN_1 reaction. 05
- B What is stereochemistry of SN_1 and SN_2 reaction? Explain with any one reaction of each. 05
- C Compare Elimination Vs. Substitution Reactions. 05
- Q.3** Attempt any **TWO** of the following
- A What is the effect of solvent on SN_1 and SN_2 ? Explain with suitable examples. 05
- B Classify various Reactive intermediates of carbon with its general structure and definition of each. 05
- C Construct the structure based on their IUPAC name. 05
- 2-Chloro Propene
 - 3-Chloro -2-methyl Butene
 - 2-Hydroxy-2methyl Hexane
 - 2-Amino Propane
 - 4-Chloro-3-Nitro Butyne

SECTION-III

- Q.4** Attempt any **FOUR** of the following
- A What is hyperconjugation? Give its different types and explain with example. 05
- B Explain orientation and reactivity of Electrophilic addition reactions in detail. 05
- C Give reaction and mechanism for Hofmann Degradation of amide. 05
- D Give reactions and preparations of ethers. 05
- E Explain inductive effect and resonance effect in detail. 05
- F Explain stability of carbocation in details. 05

Q.5 Attempt any **FOUR** of the following

- | | | |
|---|---|----|
| A | Give reactions and mechanism for free radical halogenations of Alkanes. | 05 |
| B | What is hybridization? Explain SP^2 hybridization. | 05 |
| C | Give reaction and mechanism for Aldol condensation. | 05 |
| D | Give preparations of alkenes. | 05 |
| E | Justify:- the order of reactivity of alcohols towards HX is $3^\circ > 2^\circ > 1^\circ > \text{CH}_3$. | 05 |
| F | Give preparation and reaction of Alkyl halide. | 05 |
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